Lizhong Hu

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Information Retrieval · Verifiable RAG · Thesis Evaluation

Profile

Software Engineering undergraduate in a Sino-American 4+0 dual-degree program (expected graduation: 2027). Interested in graduate study in computing, with a focus on information retrieval, verifiable RAG, and reliable software systems. Experience in thesis development, backend projects, and competitive programming.

Education

Quanzhou University of Information Engineering *2023–2027 (expected)*
Software Engineering (Sino-foreign Cooperation) GPA: 3.10/4.00

Slippery Rock University
Information Systems, BS (in progress; expected 2027) GPA: 3.583/4.000

Sino-American 4+0 dual-degree program conducted in China; approved by the Ministry of Education of China.

Research Interests

Algorithms and data structures; software and distributed systems; information retrieval and retrieval-augmented generation; reliable intelligent information systems.

Thesis & Related Projects

NoteLLM: Personal Learning QA with Verifiable RAG Ongoing
Undergraduate thesis

- Built a FastAPI backend with async SQLAlchemy, PostgreSQL, and pgvector for cosine-similarity retrieval; implemented page-aware extraction and overlapping chunks for PDF, Markdown, and TXT.
- Validated citations against retrieved chunks on the server and stored source excerpts for verification.
- A single baseline on 7 synthetic documents and 34 fixed questions recorded 100% source-level Recall@5, 97.1% automatic citation-to-source match, and 339 ms mean retrieval latency. These measure retrieval and citation-source matching on that synthetic set, not answer accuracy or general-domain performance.
- Public code and evaluation: github.com/au1bhi/NoteLLM; results in docs/evaluation/latest-results.md.

ETIPlus: Community Governance & WeChat Management Backend Mar–Apr 2025

- Built Flask APIs with PostgreSQL and MongoDB for credit scoring, anomaly audits, and group management; ran Postman regression checks.

Algorithms & Community Contributions

Matiji NOI-Pre Problem Bank | Volunteer Problem Tester Jul–Oct 2024

- Created boundary test cases, validated solutions through targeted submissions, and reported defects in test data and judge results for the Matiji/CCF problem bank. Member showcase

6th College Algorithm Design & Programming Challenge (Spring) Spring 2025
One of the problem setters

- As one of the problem setters, contributed to problem design, solution validation, difficulty calibration, and editorial review.

University ICPC/CCPC Training Team | Coordinator Sep 2024–Sep 2026

- Coordinated university ICPC/CCPC team recruitment, training, selection, and participation logistics.
- Supported university teams at the 2026 ICPC Nanchang Invitational; 2025 CCPC Nanchang Invitational/Jiangxi Provincial Contest; 2026 CCPC Fujian Invitational/Provincial Contest; 2025 ICPC Asia EC and CCPC online contests; 2026 CCPC Guizhou Provincial Contest/National Invitational; 2025 ICPC Guizhou Provincial Contest; and 2024 ICPC Kunming Regional Contest.

Campus Programming Contest Organization & Problem Setting

- ISCPC-AC Cup collegiate programming contest: principal organizer and problem setter.
- 2025 QUIE-Software School ACM Algorithm Contest (2nd Aipin Huiying Science & Technology Festival): one of the organizers and problem setters.

Competitive Programming & Awards

- 2026 National Invitational of CCPC (Fujian), 13th Fujian Collegiate Programming Contest: First to Solve Award (Problem A; problem-level award).
- The 2024 ICPC Asia Kunming Regional Contest: Honorable Medal.
- 15th Lanqiao Cup: First Prize, Fujian Provincial Round, Python Programming (University Group B), 2024.

Technical Skills

Languages & algorithms: C++, Python, Java, JavaScript, Go; algorithms and data structures.
Backend & databases: FastAPI, Flask, RuoYi, PostgreSQL, MongoDB, MySQL; REST API design and integration.
Retrieval & AI systems: RAG, LLM API integration, prompt design.
Systems & operations: Linux, Docker, Git, DOMjudge, Cloudflare Workers/CDN/DNS, WARP egress routing.
Contest tools: Familiar with Polygon (Codeforces) and Nowcoder; problem preparation, test-data management, and solution verification.